

Next-generation earthquake and fault monitoring

Earthquakes – the vibrations of the ground caused by the sudden movement of large crustal blocks on either side of a fault – are among the deadliest and most destructive geohazards. Many large metropolitan areas are located near major active faults and therefore have suffered numerous fatalities as well as consequential infrastructure and economic losses. As a sinister reminder, the recent 2023 magnitude 7.8 Kahramanmaraş earthquake killed about 60,000 people between Turkey and Syria, and the 2025 magnitude 7.7 Myanmar earthquake, more than 5,000. In the 21st century, just 10 earthquakes are responsible for more than one trillion US dollar losses worldwide.

Despite the impact of earthquakes on our societies, we know less about the mechanics of this natural phenomenon than that of atomic interactions. Our modern understanding of the earthquake phenomenon was born shortly after the great 1906 San Francisco earthquake when Reid theorized that earthquakes reflect the cycle of loading and unloading of tectonic forces along faults (Reid, 1910). Decades of digital seismic and geodetic instrumentation have then enabled the study of fast and slow tectonic deformation in Earth's crust, resulting in a solid quantitative basis for earthquake physics (Aki and Richards, 2002). Thus, nowadays, we are well capable of explaining the occurrence of large earthquakes after the fact but, at the moment, we cannot forecast the state of major faults on any useful time scale. Fundamental challenges to modeling the behavior of faults over long time scales reside in the complex, non-linear dynamics of the rocks composing Earth's crust involving physics from the scale of minerals (10^{-6} m) to the scale of large faults (10^5 m).

Nevertheless, important discoveries about earthquake physics and fault mechanics have emerged in the past two decades, driven by new observations made possible by the growing global geophysical instrumentation. Moreover, advancements in artificial intelligence (AI) in geophysics have greatly improved our ability to extract information about fault dynamics. In particular, AI-based techniques have revolutionized the detection of weak signals in continuous seismic records, thus enabling the widespread study of small magnitude earthquakes and the creation of much larger, “enhanced” earthquake catalogs. This new generation of earthquake catalogs reveals small-scale deformation on and near faults across time, thus providing new constraints on earthquake physics and fault mechanics. In my work, I have explored a new framework applied to enhanced catalogs to monitor the temporal evolution of fault properties, which opens an exciting line of research with potential implications for physics-based forecasting of large earthquakes (Beaucé et al., 2023, submitted).

In my research, I aim for a series of observational breakthroughs to address the question: **How, where and when do large earthquakes happen?** In addition to contributing to the ongoing effort to improving earthquake detection, I seek novel ways to constrain earthquake physics and use unsupervised AI to search for hidden signals of fault evolution. I therefore classify my research into three axes (see Figure 1), each of them involving significant software development:

1. Enhanced earthquake catalogs: Improving the detection and location of small earthquakes.
2. Innovative approaches to constrain earthquake physics: Establishing novel analyses of catalogs.
3. Unsupervised AI search for hidden signals of fault evolution.

1. Enhanced earthquake catalogs. Cutting-edge detection techniques including AI and other seismic network processing techniques have significantly lowered the earthquake detection threshold, allowing us to build enhanced earthquake catalogs. As a PhD student and postdoctoral researcher, I have leveraged these techniques to develop a comprehensive workflow for earthquake detection and location, enabling me to overcome traditional noise limitations and characterize small-magnitude seismicity across diverse tectonic settings (Beaucé et al., 2019, 2022, 2023, 2025). I built the workflow upon highly efficient computer programs accelerated by graphic processing units (GPUs) to achieve great computation speeds and enable the analysis of large datasets (Beaucé et al., 2018, 2024). I have thus become a leader in open-source community tools for earthquake detection. This workflow has initially been developed for seismic recordings but can naturally be extended to fiber optics ground deformation measurements (distributed acoustic sensing).

2. Innovative approaches to constrain earthquake physics. The large number of small-magnitude events in enhanced earthquake catalogs provides rich information on small-scale deformation on and near faults that is lacking in conventional catalogs. These new massive data products must be analyzed in novel ways to lead to observational breakthroughs that provide new constraints on earthquake physics. I have been leading the development of a new methodology to constrain the physical state of a fault over time, which consists in analyzing the subtle influence of tidal stresses – that is, stresses caused by the gravitational forces of the Moon and the Sun – on the occurrence of small earthquakes on and near faults (see complementary details in the Scientific

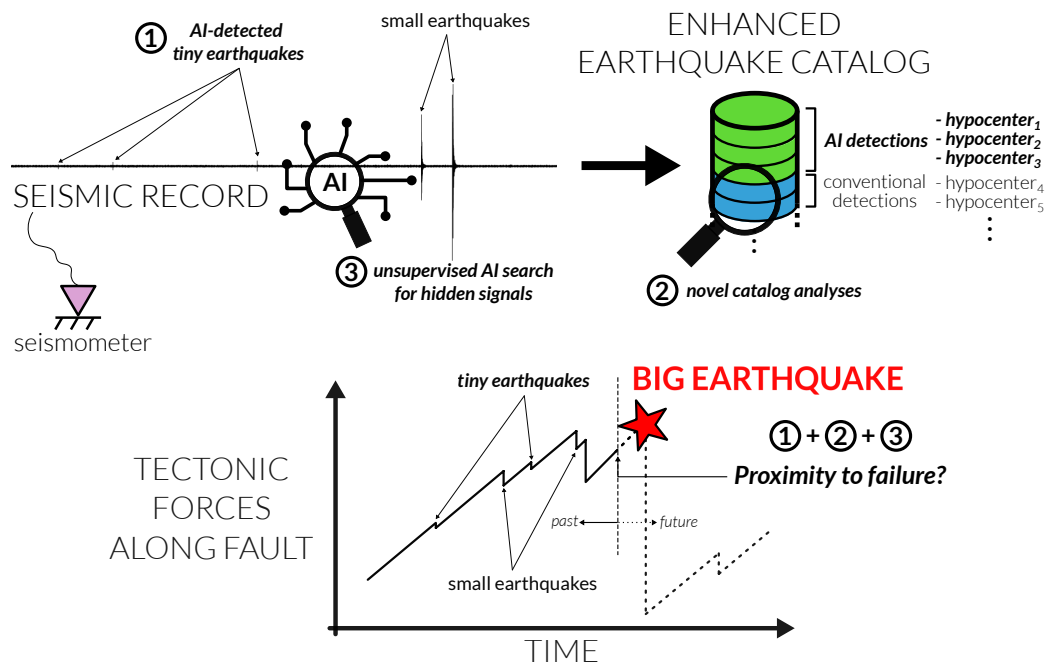


Figure 1. Monitoring the evolution of a fault toward the high stress state necessary for a large earthquake using 1) AI-based detection, 2) novel catalog analyses for new physical constraints and 3) unsupervised AI search for hidden signals.

Vision Statement). I have shown that this tidal modulation of seismicity became significantly stronger along the Ridgecrest fault in Southern California in the few years preceding the 2019 magnitude 7.1 earthquake, and weak afterward (Beaucé et al., 2023, submitted). These observations provide new constraints to improve our understanding of fault frictional behavior, a central piece of earthquake physics, and better understand the processes leading up to large earthquakes. This work presents major implications for the future of physics-based earthquake forecasting, and I will therefore comprehensively extend this research to other major faults.

3. Unsupervised AI search of hidden signals. While earthquake catalogs provide valuable information on seismicity, they offer only a highly simplified view of the underlying continuous seismic records, ignoring a significant portion of the seismic wavefield. Indeed, sub-detection seismic sources reveal the continuous but quiet crustal deformation processes and the time-dependent characteristics of seismic “noise” reflect the evolving properties of the crust. The complexity of the seismic wavefield eludes, for now, our physical models, but an empirical search for structured patterns using unsupervised AI can potentially uncover hidden signals of fault evolution. To achieve this goal, I will build on cutting-edge AI exploration techniques developed within my research group (Steinmann et al., 2022) and recent advances in foundation models.

References

- Keiiti Aki and Paul Richards. *Quantitative seismology*. MIT Press, 2002.
- Eric Beaucé, William B Frank, and Alexey Romanenko. Fast matched filter (FMF): An efficient seismic matched-filter search for both CPU and GPU architectures. *Seismological Research Letters*, 2018.
- Eric Beaucé, William B Frank, Anne Paul, Michel Campillo, and Robert D van Der Hilst. Systematic detection of clustered seismicity beneath the Southwestern Alps. *Journal of Geophysical Research: Solid Earth*, 2019.
- Eric Beaucé, Robert D van der Hilst, and Michel Campillo. Microseismic constraints on the mechanical state of the North Anatolian fault zone 13 years after the 1999 M7.4 Izmit earthquake. *Journal of Geophysical Research: Solid Earth*, 2022.
- Eric Beaucé, Piero Poli, Felix Waldhauser, Benjamin Holtzman, and Christopher Scholz. Enhanced tidal sensitivity of seismicity before the 2019 magnitude 7.1 Ridgecrest, California earthquake. *Geophysical Research Letters*, 2023.
- Eric Beaucé, William B Frank, Léonard Seydoux, Piero Poli, Nathan Groebner, Robert D van der Hilst, and Michel Campillo. BPMF: A Backprojection and Matched-Filtering Workflow for Automated Earthquake Detection and Location. *Seismological Research Letters*, 2024.
- Eric Beaucé, Felix Waldhauser, David Schaff, Won-Young Kim, and Folarin Kolawole. The 5 April 2024 Mw 4.8 Tewksbury, New Jersey Aftershock Sequence Resolved With Machine-Learning-Enhanced Detection Methods. *Geophysical Research Letters*, 2025.
- Eric Beaucé, Piero Poli, Felix Waldhauser, Benjamin Holtzman, and Christopher Scholz. Shift and increase in tidal modulation of seismicity in the Ridgecrest fault zone before and after the 2019 magnitude 7.1 earthquake. *Bulletin of the Seismological Society of America*, submitted.
- Harry Fielding Reid. The mechanics of the earthquake. *The California Earthquake of April 18, 1906, Report of the State Earthquake Investigation Commission*, 1910.
- Rene Steinmann, Leonard Seydoux, Eric Beaucé, and Michel Campillo. Hierarchical exploration of continuous seismograms with unsupervised learning. *Journal of Geophysical Research: Solid Earth*, 2022.